

Practical Assignment 2 - Constructing the Bacterial Tree of Life

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Part I- Data collection

To construct phylogenetic trees, data was collected for the strains mentioned in Table 1 (Archaea) and Table 2 (Bacteria).

Domain	Species
Archaea	Acidianus ambivalens DSM 3772
	Halobacterium salinarum 91-R6
	Sulfolobus acidocaldarius 98-3

Table 1: Archaea species considered (Outgroup)

Domain	Phylum	Species
Bacteria	Proteobacteria (alpha)	Acetobacter aceti L 40
	Proteobacteria (beta)	Achromobacter denitrificans 12
	Proteobacteria (gamma)	Acidithiobacillus ferrooxidans TM
	Firmicutes	Acidaminococcus fermentans VR4
	Firmicutes	Phascolarctobacterium faecium QUM 3679
	Firmicutes	Enterococcus florum Gos25-1
	Firmicutes	Vallitalea guaymasensis L81
	Firmicutes	Alkalibacterium pelagium T143-1-1
	Firmicutes	Staphylococcus aureus 533 R4
	Firmicutes	Enterococcus durans 98 D
	Actinobacteria	Winkia neuui CIP 102477
	Actinobacteria	Thermopolyspora (Nonomuraea) flexuosa 435

	Actinobacteria	Ferrimicrobium acidiphilum T23
	Actinobacteria	Kocuria palustris TAGA27
	Actinobacteria	Saccharopolyspora rectivirgula DSM 43747
	Cyanobacteriota	Nostoc entophytum Nostoc sp. C1.8
	Cyanobacteriota	Coleofasciculus chthonoplastes VOE
	Spirochaetes	Leptospira inadai 10
	Spirochaetes	Spirochaeta thermophila Z-1203
	Chlamydiae	Simkania negevensis Z
	Chlamydiae	Waddlia chondrophila 2032/99
	Chloroflexi	Anaerolinea thermolimosa IMO-1
	Chloroflexi	Bellilinea caldifistulae GOMI-1
	Synergistetes	Acetomicrobium flavidum P
	Synergistetes	Lactivibrio alcoholicus 7WAY-8-7
	Deinococcota	Deinococcus xinjiangensis X-82
	Deinococcota	Calidithermus roseus DSM 29972
	Bacteroidetes	Capnocytophaga cynodegmi 79120329
	Bacteroidetes	Bacteroides pyogenes P39-88
	Bacteroidetes	Lacinutrix algicola AKS 293
	Bacteroidetes	Bacteroides fragilis EN-2

Table 2: Bacteria strains considered for tree construction (n=31)

As required by the assignment, Table 2 considers a diverse collection of Bacteria from various phyla, like Firmicutes, Actinobacteria, Chloroflexi and taxa like Proteobacteria. I've included three archaea to ensure a stable outgroup. All trees constructed were rooted in Archaea.

The 16S rRNA sequences for each species was primarily collected from SILVA, though NCBI was considered for a few species. For phenotype matrix, the following features were initially considered: Habitat (Mesophile or Thermophile only), Metabolism (Aerobic, Anaerobic, Photosynthetic), Risk Group (German) as a measure of pathogenicity, Morphology: (e.g., Rod-shaped, Coccus-shaped), Motility, Gram staining (+ve/-ve or Gram-variable). For these, BacDive and DSMZ were primary resources, though literature was considered for some underdocumented species.

However, constructing UPGMA-based dendrograms using these features alone meant that Archaea weren't each other's closest neighbours, making it difficult to root. There were 2 ways to circumvent this: i) Removing 2 Archaea from the database, so there is an unequivocal outgroup, ii) Consider additional morphological characteristics that largely separate Archaea from Bacteria for a new Phenotype tree. For apt comparison and to avoid loss of information, additional phenotypes considered. The new features were the presence of peptidoglycan in the cell wall (mostly present in all bacteria, absent in all Archaea), ether-linked isoprenoid lipids and the presence of S-layer dominant cell walls (present in Archaea, absent in Bacteria). I consider this morphology tree in my analysis.

I one-hot encoded the entire phenotype dataframe for a binary output, which lead to my final phenotype matrix:

Species	Obligate	Aerobe	Anaerobe	Intracellular	Photosynthetic	Mesophilic	Motile	RG1	Gram_pos	Gram_variable	Morph_rod	Morph_spherical	Morph_ovoid	Morph_spiral	Morph_filamentous	Morph_other	Peptidoglycan	Ether_Linked_lipids	S_layer_Cell_wall
Acetobacter aceti	1	1	0	0	0	1	0	1	0	0	1	0	0	0	0	0	1	0	0
Acetomicrobium flavidum	1	0	1	0	0	0	1	1	0	1	1	0	0	0	0	0	1	0	0
Achromobacter denitrificans	0	0	1	0	0	1	1	0	0	1	0	0	0	0	0	0	1	0	0
Acidaminococcus fermentans	1	0	1	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0
Acidianus ambivalens	0	0	1	0	0	0	0	1	0	0	0	1	0	0	0	0	0	1	1
Acidithiobacillus ferrooxidans	0	0	1	0	0	1	1	1	0	0	1	0	0	0	0	0	1	0	0
Alkalibacterium pelagium	0	0	0	0	0	1	1	1	1	0	1	0	0	0	0	0	1	0	0
Anaerolinea thermolimosa	1	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0
Bacteroides fragilis	1	0	1	0	0	1	0	0	0	0	1	0	0	0	0	0	1	0	0
Bacteroides pyogenes	1	0	1	0	0	1	0	0	0	0	1	0	0	0	0	0	1	0	0
Bellilinea caldifutulae	1	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0
Calditerrus roseus	1	1	0	0	0	0	0	1	0	0	1	0	0	0	0	0	1	0	0
Capnocytophaga cynodegmi	0	0	1	0	0	1	1	0	0	0	1	0	0	0	0	0	1	0	0
Coleofasciculus chthonoplastes	0	0	0	0	1	1	1	1	0	0	0	0	0	0	1	0	1	0	0
Deinococcus xinjiangensis	0	0	1	0	0	1	0	1	1	0	1	0	0	0	0	0	1	0	0
Enterococcus durans	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	0	1	0	0
Enterococcus faecium	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	0	1	0	0
Ferrirobium acidiphilum	0	0	1	0	0	1	1	1	0	0	1	0	0	0	0	0	1	0	0
Halobacterium salinarum	1	1	0	0	0	1	0	1	0	0	1	0	0	0	0	0	0	1	1
Kocuria palustris	1	1	0	0	0	1	0	1	1	0	0	1	0	0	0	0	1	0	0
Lacinutrix algicola	1	1	0	0	0	1	1	1	0	0	1	0	0	0	0	0	1	0	0
Lactivibrio alcoholicus	1	0	1	0	0	1	0	1	0	0	1	0	0	0	0	0	1	0	0
Leptospira inadai	1	1	0	0	0	1	1	0	0	0	0	0	0	1	0	0	1	0	0
Nostoc sp.	0	0	0	0	1	1	1	1	0	0	0	1	0	0	0	0	1	0	0
Phascolarctobacterium faecium	1	0	1	0	0	1	0	1	0	0	1	0	0	0	0	0	1	0	0
Saccharopolyspora rectivirgula	1	1	0	0	0	0	0	1	1	0	1	0	0	0	0	0	1	0	0
Sinkania negevensis	1	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0
Spirochaeta thermophila	1	0	1	0	0	0	1	1	1	0	0	0	0	1	0	0	1	0	0
Staphylococcus aureus	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	0	1	0	0
Sulfolobus acidocaldarius	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	1	1
Thermoplasma (Nonomuraea) flexuosa	1	1	0	0	0	1	0	1	1	0	0	0	0	0	1	0	1	0	0
Valitalea guaymasensis	1	0	1	0	0	0	1	1	1	0	1	0	0	0	0	0	1	0	0
Waddlia chondrophila	1	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0
Winkia neui	0	0	1	0	0	1	0	0	1	0	1	0	0	0	0	0	1	0	0

Fig 1: Final Binary Phenotype Matrix

Part 2: Tree Construction

Phenotype-based Dendrogram

The Phenotype matrix in Fig 1 was converted into a fasta-like format using a python script for input into DendroUPGMA. Jaccard Distance was used as the distance matrix, with 100 bootstraps. Other parameters can be seen in Fig 2.



DendroUPGMA: A dendrogram construction utility.
Created at March 2002.
By Santi Garcia-Valle (santi.garcia-valle AT urv.net) and Pere Puigbo
Biochemistry and Biotechnology Department,
Universitat Rovira i Virgili (URV), Tarragona, Spain.
[Version history](#) [Tutorial](#) [Examples](#) [Server use](#)

Use this program to create a dendrogram from (a) sets of variables, (b) a similarity matrix or (c) a distance matrix. The program calculates a similarity matrix (only for option a), transforms similarity coefficients into distances and makes a clustering using the Unweighted Pair Group Method with Arithmetic mean (UPGMA) or Weighted Pair Group Method with Arithmetic Mean (WPGMA) algorithm.

For Option (a) -Construct a dendrogram from a set of variables- two input formats are available:

- Format 1: fasta-like format - Two lines, the first one must begin with the ">" character (this will be the identifier line) and the second one that contains the variables (separated by tabs or spaces, not a combination of both).
- Format 2: Only one line for each set of variables (separated by spaces). The first one will be assumed to be the identifier.

For Options (b) and (c) -Construct a dendrogram from a similarity or distance matrix- the input must be a similarity or distance matrix in CSV or tab-delimited format.

```
>Phascolarctobacterium_faecium
1 0 0 0 0 1 1 1 1 0 0 0 1 0 0 0 0 0 1 0 0
>Saccharopolyspora_rectivirgula
1 0 1 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 0 0
>Sinkania_negevensis
1 0 0 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0
>Spirochaeta_thermophila
1 0 1 0 0 0 0 1 1 1 0 0 0 0 1 0 0 0 1 0 0
>Staphylococcus_aureus
0 0 0 1 0 0 1 0 0 1 0 0 1 0 0 0 0 0 1 0 0
>Sulfolobus_acidocaldarius
1 1 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 1 0
>Thermopolyspora_(Nonomuraea)_flexuosa
1 1 0 0 0 1 0 1 1 0 0 0 0 0 0 0 1 0 1 0 0
>Vallitalea_guaymasensis
1 0 1 0 0 0 0 1 1 1 0 1 0 0 0 0 0 0 1 0 0
>Waddlia_chondrophila
1 0 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0
>Winkia_neuui
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

Choose the input:

☒ (a) sets of variables (raw data)

Choose the similarity index or distance coefficient used to compare between the set of variables (Use Jaccard or Dice for binary data):

Similarity index: ☐ Pearson correlation coefficient (r) ☒ Jaccard index (Tanimoto) ☐ Dice coefficient

Distance coefficient: ☐ Euclidean distance ☐ Manhattan distance (city block or taxicab distance) ☐ Mean square deviation (MSD) ☐ Root Mean Square Deviation (RMSD)

Do you want to normalize (standardize) the input data? (only available for distance coefficients) ☒ No ☐ Yes

If the Pearson coefficient (r) has been chosen, choose the transformation of r to generate the distance values (d):

☒ $d = (1 - r) * 100$ ☐ $d = -\ln(\text{abs}(r))$

Do you want to generate 100 bootstrap replicates? ☐ No ☒ Yes

Do you want to omit rows with identical values for all the variables? ☒ No ☐ Yes

☐ (b) a similarity matrix

Do you want to omit rows with a similarity of 1? ☐ No ☐ Yes

☐ (c) a distance matrix

Choose the clustering method:

☒ UPGMA (Unweighted Pair Group Method with Arithmetic Mean) ☐ WPGMA (Weighted Pair Group Method with Arithmetic Mean)

Fig 2: Image showing job submission and parameters chosen for morphology tree construction

The Newick string outputted by DendroUPGMA was visualized using the iTOL web-server. The tree was rooted in the pertinent outgroup of Archaea, yielding the tree seen in Fig 3:

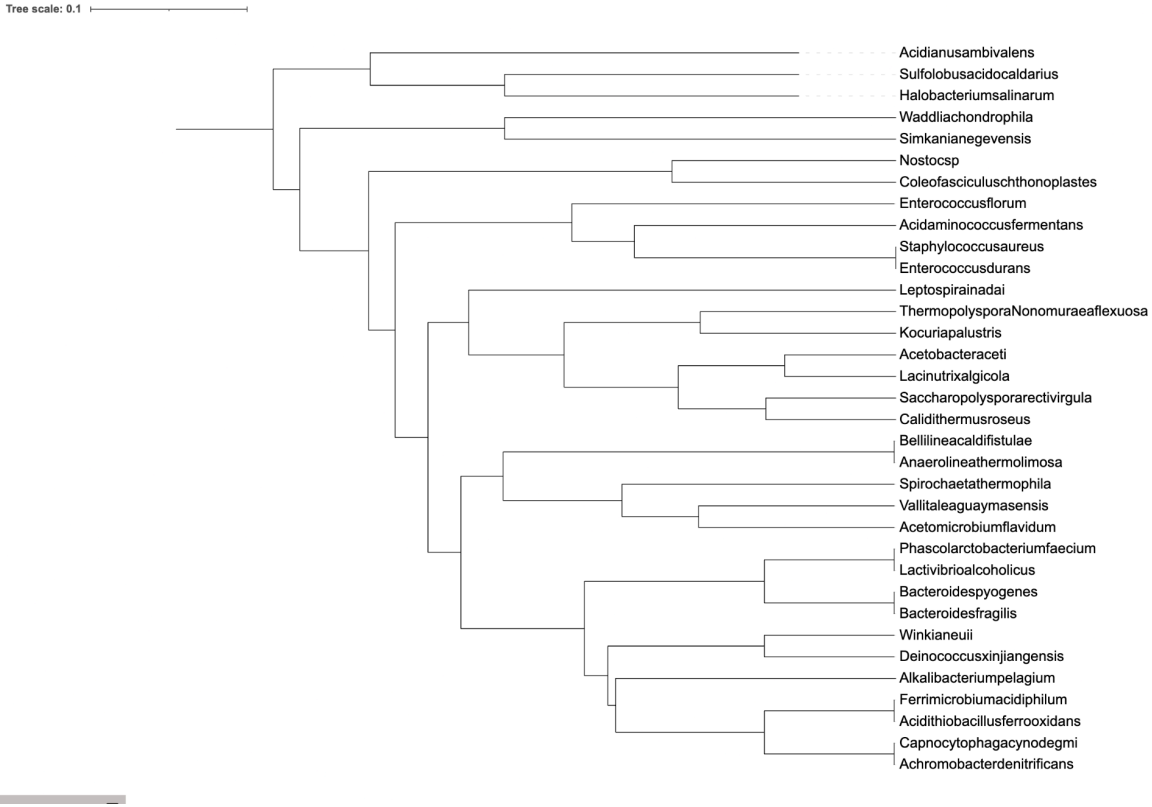


Fig 3: Morphology Tree (rooted)

Molecular Tree Construction

For Multiple Sequence Alignment of the 16S rRNA sequences, I used Clustal Omega's web-server. The clustal-num file served as the input for all the molecular trees constructed here. The molecular trees for UPGMA and NJ were constructed using EMBL-EBI's 'Simple Phylogeny'. The parameters chosen for UPGMA and NJ are shown in Fig 4 and Fig 5 respectively:

Simple Phylogeny

Phylogeny

[Job Dispatcher](#)
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[Your Jobs](#)
[Input form](#)

Feedback

This tool provides access to phylogenetic tree generation methods from the ClustalW2 package. Please note this is NOT a multiple sequence alignment tool. To perform a multiple sequence alignment please use one of our MSA tools.

Input sequence

Paste your sequence here - or use the example sequence

Choose file

simplified_ms...-clustal_num

Use the example

Clear sequence

More example inputs

Parameters

TREE FORMAT ①

DISTANCE CORRECTION ②

EXCLUDE GAPS ③

CLUSTERING METHODS ④

PERCENT IDENTITY MATRIX ⑤

Phylip

OFF

OFF

UPGMA

OFF

Submit

Title

UPGMA_MSA_A3

Submit

Fig 4: UPGMA Job submission

Simple Phylogeny

Phylogeny

[Job Dispatcher](#)
[Help & Privacy](#)
[Your Jobs](#)
[Input form](#)

Feedback

This tool provides access to phylogenetic tree generation methods from the ClustalW2 package. Please note this is NOT a multiple sequence alignment tool. To perform a multiple sequence alignment please use one of our MSA tools.

Input sequence

Paste your sequence here - or use the example sequence

Choose file

simplified_ms...-clustal_num

Use the example

Clear sequence

More example inputs

Parameters

TREE FORMAT ①

DISTANCE CORRECTION ②

EXCLUDE GAPS ③

CLUSTERING METHODS ④

PERCENT IDENTITY MATRIX ⑤

Phylip

OFF

OFF

Neighbour-joining

OFF

Submit

Title

NJ_MSA_A3

Submit

Fig 5: NJ Job submission

The outputs from UPGMA and NJ runs were visualised using EMBL-EBI's Phylogram tool. Additionally, iTOL web-server was also used during analysis, with the .tree file being used as an input. The trees visualised using Phylogram are attached here (after rooting):

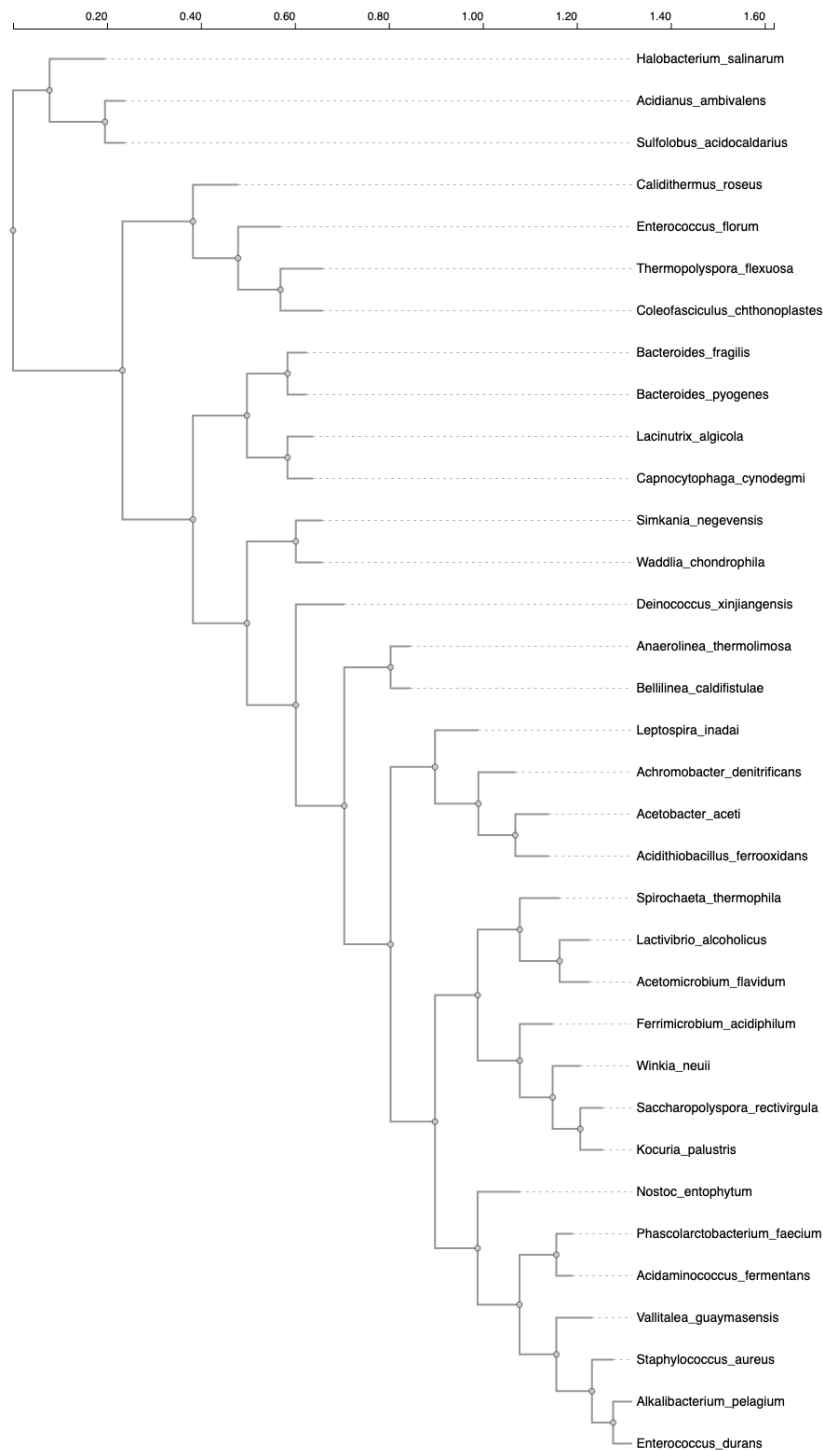


Fig 6: UPGMA Tree

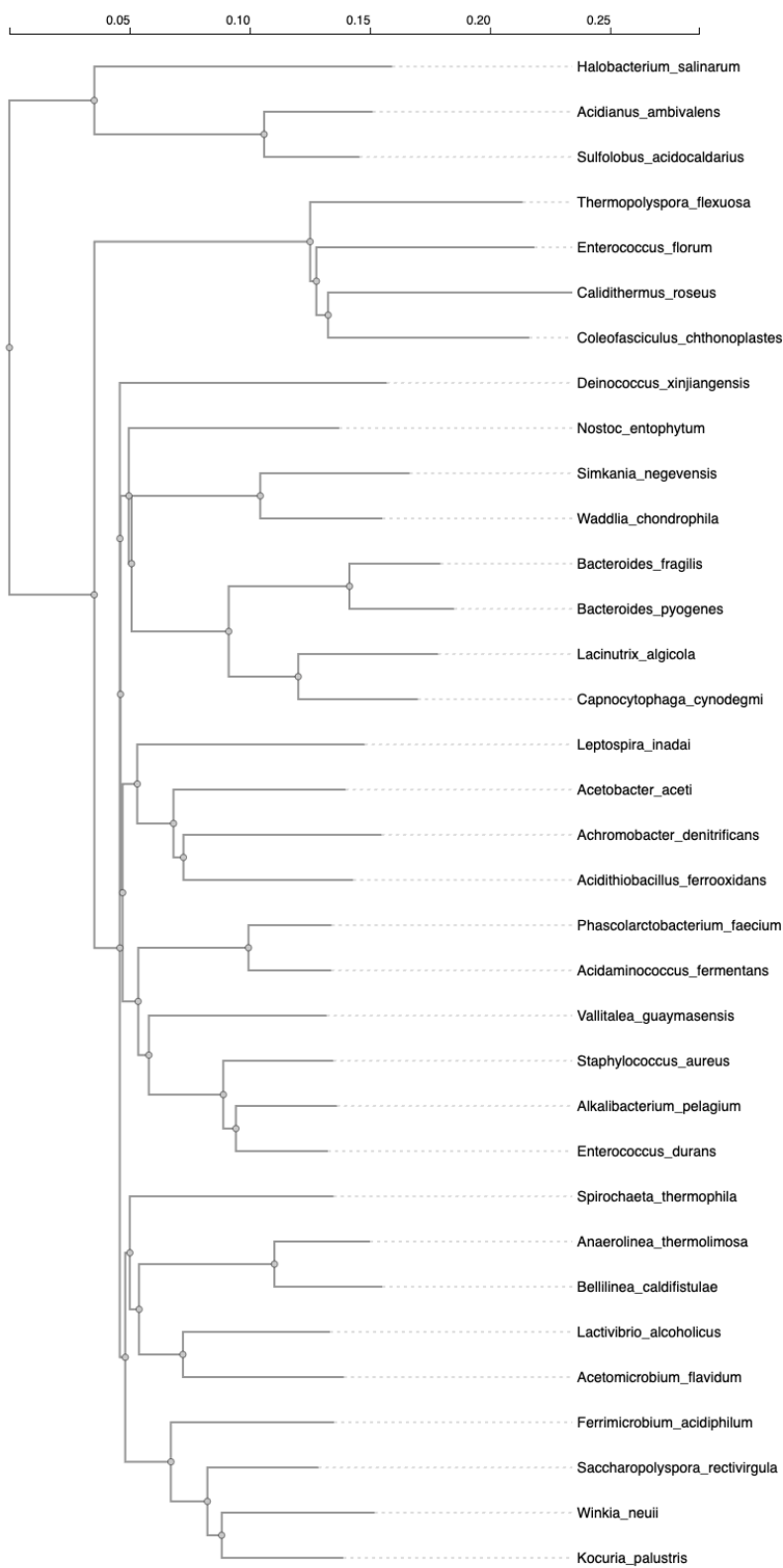


Fig 7: NJ Tree

Maximum Likelihood Tree

For this, the MSA output was inputted to IQTREE. The job submission and parameters are visible in Fig 8.

IQ-TREE web server: fast and accurate phylogenetic trees under maximum likelihood

Server load: 10%Trifinopoulos J, Nguyen LT, von Haeseler A, Minh BQ (2016) *Nucl. Acids Res.* 44 (W1): W232-W235. doi: 10.1093/nar/gkw256

Tree Inference | Model Selection | Analysis Results

Alignment file : C:\fakepath\clustalo-I20250819-1

Use example alignment: ☐ Yes

Sequence type: ☐ Auto-detect ☒ DNA ☐ Protein ☐ Codon
☐ DNA->AA ☐ Binary ☐ Morphology

Partition file: This field is optional.

Partition type: ☒ Edge-linked ☐ Edge-unlinked

Substitution Model Options

Substitution model: Auto

FreeRate heterogeneity: ☐ Yes [+R]

Rate heterogeneity: ☐ Gamma [+G] ☐ Invar. sites [+I]

#rate categories: 4

State frequency: ☒ Empirical (from data) ☐ AA model (from matrix) ☐ ML-optimized
☐ Codon F1x4 ☐ Codon F3x4

Ascertainment bias correction: ☐ Yes [+ASC]

Branch Support Analysis

Bootstrap analysis: ☐ None ☒ Ultrafast ☐ Standard

Number of bootstrap alignments: 1000

Create .ufboot file: ☐ Yes (write bootstrap trees to .ufboot file)

Maximum iterations: 1000

Minimum correlation coefficient: 0.99

Single branch tests:

SH-aLRT branch test: ☐ No ☒ Yes #replicates: 1000

Approximate Bayes test: ☐ Yes

IQ-TREE Search Parameters

Perturbation strength: 0.5

IQ-TREE stopping rule: 100

Fig 8: IQTree Job Submission and parameters

This was visualised using iTOL, and yields the following Tree:

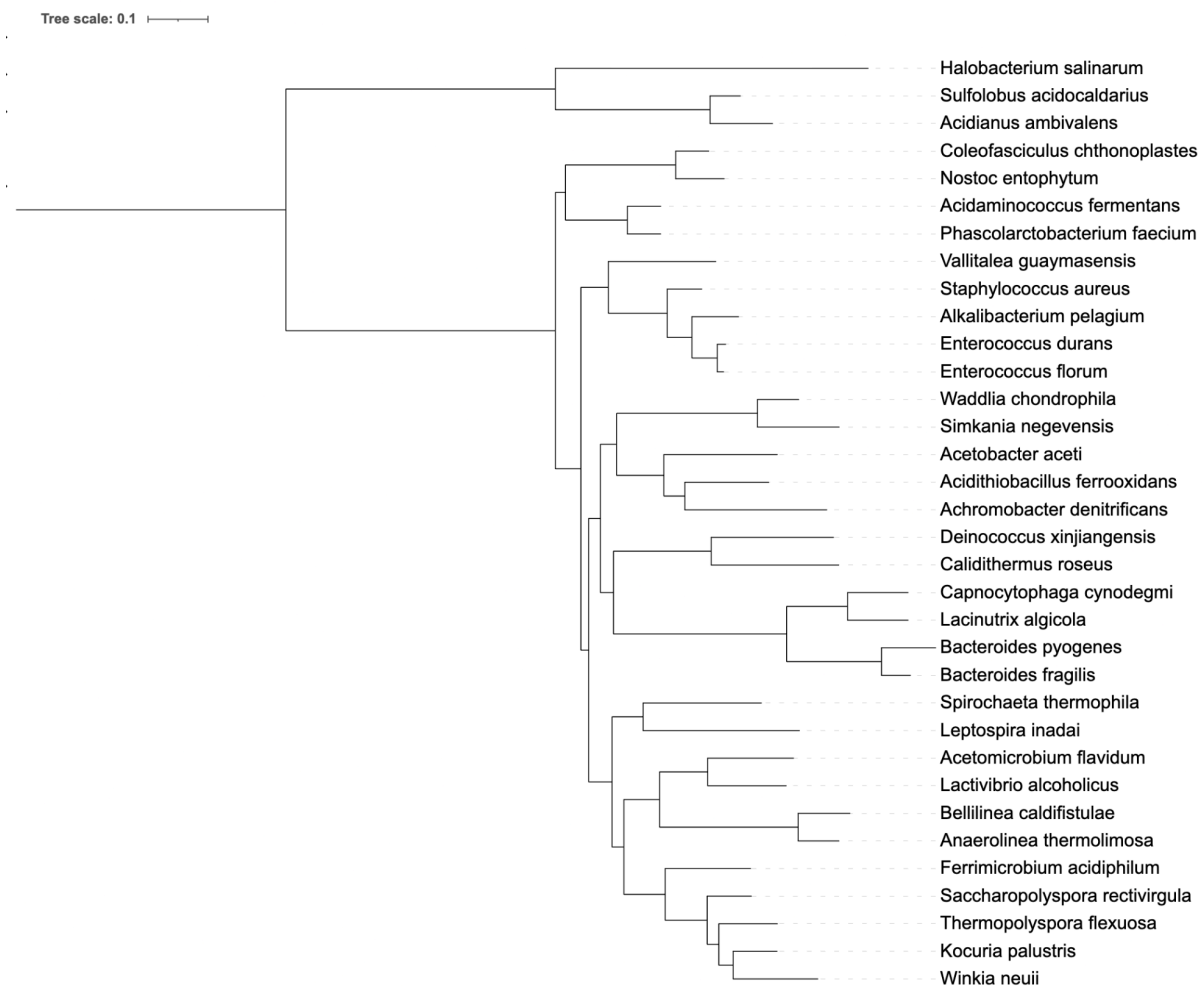


Fig 9: Maximum Likelihood Tree

Part 3: Topology Analysis of Molecular and Phenotype Trees

Comparing phenotype-based dendrogram to your 16S rRNA trees.

Across all reconstructions, individuals from Chlamydiae (*Simkania*, *Waddlia*) form a close monophyletic relationship. Similarly, members of Chloroflexi (*Anaerolinea*, *Bellilinea*) are also assigned to be each other's closest relative. These agreements between the morphology tree and all three molecular trees show that the morphological features do capture evolutionary relationships in a way consistent with molecular methods. Also, Archaea is monophyletic, and

can be reliably assigned to be outgroup across all analyses, though their position relative to Bacteria varies slightly between methods. *Bacteroides fragilis* and *Bacteroides pyogenes* form a pair in morphology, NJ, & UPGMA trees, though ML separates them slightly.

Further, the phenotypic tree doesn't assign Proteobacteria members (*Achromobacter*, *Acidithiobacillus*, *Acetobacter*) to be close to each other, though the molecular methods do. Similarly, the Negativicute pair of *Phascolarctobacterium* and *Acidaminococcus* (from Firmicutes), are consistently joined in ML, NJ, and UPGMA, but morphology instead links each with unrelated partners. An identical trend is observed with *Synergistetes* as Morphology splits them (*Acetomicrobium* paired with *Vallitalea*, *Lactivibrio* with *Phascolarctobacterium*), while ML, NJ, and UPGMA recover them as sisters.

Morphology (with UPGMA and NJ) misses/misplaces several well-supported groupings that ML rightly assigns. ML groups Deinococcota (*Deinococcus* and *Calidithermus*), Spirochaetes (*Leptospira* and *Spirochaeta*) closely, while also accurately grouping members of the *Enterococcus*. The morphology tree consistently breaks these apart, sometimes grouping *Deinococcus* with Actinobacteria (Winkia) or *Calidithermus* with Saccharopolyspora. NJ and UPGMA similarly fail to reproduce these groupings.

Topologies comparison of the three molecular trees

As the section above also touches upon the differences between performance of the three molecular trees, for clarity, I give a brief comparison among the three molecular trees (rooted in Archaea) here. All three trees recover Chlamydiae, Chloroflexi, Synergistetes, and the Negativicute pair. They also cluster the Proteobacteria trio. There are some differences between UPGMA and NJ trees. Cyanobacteria were split differently (NJ places *Coleofasciculus* with *Calidithermus* while UPGMA pairs *Coleofasciculus* with *Thermopolyspora*). Also, NJ assigns *E. florum* to be close to the Cyanobacteria/Deinococcota area, while UPGMA drags it into an Actino/Cyano cluster.

ML uniquely restores Deinococcota, Spirochaetes, Enterococcus, and a cohesive Actinobacteria block, whereas NJ and UPGMA split these groups and also split Cyanobacteria (placing *Coleofasciculus* with *Calidithermus* in NJ and with *Thermopolyspora* in UPGMA). This qualitatively observed robustness of ML is also why we can consider it to be the best tree-construction method in terms of inferring evolutionary history. However, for mathematical precision, there is a need for a method to quantify how well a tree assigns relationships. To do so, we need to split the clades into groups, and we can then calculate how accurate the group assignment is, based on how 'pure' the group is. For this purpose, we can use entropy-based, or gini-impurity measures. I've conducted a rough quantification using gini-impurity, which supports the qualitative observation of ML being the best construction method. I've covered this in the Supplementary part of this document for clarity. The rough quantitative part of the analysis

shows that ML is the best method, followed by UPGMA/NJ (very similar) and finally by the morphological tree.

Why Maximum Likelihood Methods are the best for Phylogenetic tree construction

We can explain why ML is the best in this case in two ways. One by considering the inherent disadvantages of UPGMA, NJ, and Morphology trees, and the other by recognizing what algorithmic nature makes ML robust.

UPGMA as a method, assumes equal evolutionary rates across lineages. As often, in real cases we study, don't fulfill this assumption, branch lengths are misestimated and taxa with fast rates are misplaced/misassigned. The NJ method relies on pairwise distances between sequences. And due to the nature of the algorithm, these distances 'collapse' the information in the alignment into a single number per pair. This means that there is a loss of information on site-specific patterns. This makes NJ vulnerable to phenomena such as long-branch attraction and hence less able to resolve deeper relationships, which could explain the performance seen in this report. These methods however are still valuable thanks to their algorithmic complexity (space / time) as compared to ML. We see that overall, qualitatively (and quantitative: below), the morphology tree is the worst performing tree in this case. This is because inferences of evolutionary history based on phenotype are inherently inaccurate because only a limited set of traits are typically considered, which cannot capture all the changes that evolution brings about. Phenomena like homoplasy (as in convergent evolution, evolutionary reversal and parallel evolution) further confounds inference, as unrelated taxa (that are far based on sequence distance) may share similar phenotypes. Similarly, deep genetic divergence may leave morphology unchanged, or small genetic distances can translate into disproportionately large phenotypic shifts. These phenomena make phenotype-only trees unreliable compared to molecular phylogenies.

ML however evaluates full sequence likelihoods under explicit substitution models (like say Jukes-Cantor, K2P or more advanced ones), not just distances. This allows it to use all available information, like patterns of substitutions across sites and accommodate for rate heterogeneity across both lineages & sites. Further, the ML algorithm provides a way to estimate the 'support' for each estimation (via likelihood ratios, bootstrap, etc.), making its trees both better justified and interpretable.

— ## END OF ASSIGNMENT ## —

— ## SUPPLEMENTARY SECTION TO SUPPORT ABOVE WORK ## —

Supplementary: Roughly quantifying how well a construction method performs.

Step 1: Topological splitting rule (common to all trees)

To consider how good a construction method is, given that we already know the parent taxa of the species we have considered, we need to be able to assess how consistently the tree groups close taxa together. One way we could do this is by splitting the trees into groups. However, these groups shouldn't be too large, as tight, monophyletic groups may be seen as 'impure', why too small groups, despite possibly incorrectly assigning relationships. Herewith, I have performed a somewhat arbitrary grouping, following this rough algorithm.

Root in Archaea, remove them from further consideration, then:

1. Traverse from the root until you hit a node whose children are comparably deep (not just terminal twigs).
2. Take each child's entire descendant subtree as a candidate group.
3. If that subtree contains successive shallow splits (i.e., several near-twigs adding little extra depth), collapse them together so we don't artificially inflate "purity" with micro-groups.
4. Repeat algorithm if a clade's own children are comparably deep; stop when further splits are shallow

To compare across methods, I made sure to repeat this rough heuristic till a nearly equal number of groups were obtained for each method.

Step 2: Impurity Quantification in each group

Let a phylum P have n species total. Map each species to its group ID (from the split). Suppose those n species occupy k groups with counts $c_1 \cdots c_k$ (so $\sum_{\forall i} c_i = n$).

Now, we can define a the Gini impurity of phylum P to be

$$G(P) = 1 - \sum_{i=1}^k \frac{c_i^2}{n^2}$$

Now, If all n species of P are in **one** group, $G(P)=0$. If they are evenly split across multiple groups, $G(P)$ increases toward its maximum.

Step 3: Tree Level Score

To get a unified, tree-level score, I consider a frequency weighted average for each group. Thus, the tree-level weighted average is:

$$\bar{G} = \sum_{\forall P} \frac{\# \text{ species in } P}{\text{Total species in Tree}} \times G(P)$$

Using this schema, we can calculate the weighted average for any tree. Here, I do so first the Maximum Likelihood tree in Fig 9.

With Archaea removed, we can split the Tree into the following groups, as follows

- R1: Bacteroides fragilis*
- R2: Bacteroides pyogenes*
- R3: Lacinutrix_algicola, Capnocytophaga_cynodegmi*
- R4: Deinococcus xinjiangensis, Calidithermus roseus*
- R5: Thermopolyspora flexuosa, Winkia neuui, Kocuria palustris, Saccharopolyspora rectivirgula, Ferrimicrobium acidiphilum*
- R6: Anaerolinea thermolimos, Bellilinea caldifistulae*
- R7: Lactivibrio alcoholicus, Acetomicrobium flavidum*
- R8: Leptospira inadai, Spirochaeta thermophila*
- R9: Vallitalea guaymasensis, Staphylococcus aureus, Alkalibacterium pelagium, Enterococcus florum, Enterococcus durans*
- R10: Phascolarctobacterium faecium, Acidaminococcus fermentans, Nostoc entophytum, Coleofasciculus chthonoplastes*
- R11: Achromobacter denitrificans, Acidithiobacillus ferrooxidans, Acetobacter aceti*
- R12: Simkania negevensis, Waddlia chondrophila*

Now, for each phylum P , we can find the gini impurity

- **Actinobacteria** (all in R5) $\Rightarrow G = 0$
- **Bacteroidetes** (B. fragilis $\rightarrow$ R1; B. pyogenes $\rightarrow$ R2; Lacinutrix & Capnocytophaga $\rightarrow$ R3)
Counts: $(c_{R1}, c_{R2}, c_{R3}) = (1, 1, 2)$

$$\text{Thus, } G = 1 - \left(\frac{1^2}{4^2} + \frac{1^2}{4^2} + \frac{2^2}{4^2} \right) = 0.625.$$

- **Chlamydiae** (in R12) $\Rightarrow G = 0$.
- **Cyanobacteriota** (in R10) $\Rightarrow G = 0$.
- **Deinococcota** (in R4) $\Rightarrow G = 0$
- **Firmicutes** (E. florum, E. durans, Vallitalea, Alkalibacterium, Staph $\rightarrow$ R7, Phascolarctobacterium, Acidaminococcus $\rightarrow$ R10)
Counts: $(c_{R9}, c_{R10}) = (5, 2) \Rightarrow G = 0.4082$
- **Proteobacteria** (all R9) $\rightarrow G = 0$.
- **Spirochaetes** (both R8) $\rightarrow G = 0$.
- **Synergistetes** (both R7) $\rightarrow G = 0$.

$$\text{Weighted average (ML): } \bar{G}_{ML} = \frac{4}{31} \times 0.625 + \frac{7}{31} \times 0.4082 = 0.1728$$

We can do the same for the other trees:

For NJ:

NJ groups:

G1: *Calidithermus roseus*, *Deinococcus xinjiangensis*

G2: *Thermopolyspora flexuosa*, *Coleofasciculus chthonoplastes*, *Enterococcus florum*

G3: *Simkania negevensis*, *Waddlia chondrophila*

G4: *Bacteroides fragilis*, *Bacteroides pyogenes*, *Lacinutrix algicola*, *Capnocytophaga cynodegmi*

G5: *Achromobacter denitrificans*, *Acidithiobacillus ferrooxidans*, *Acetobacter aceti*

G6: *Leptospira inadai*

G7: *Phascolarctobacterium faecium*, *Acidaminococcus fermentans*

G8: *Staphylococcus aureus*, *Alkalibacterium pelagium*, *Enterococcus durans*

G9: *Vallitalea guaymasensis*

G10: *Anaerolinea thermolimosa*, *Bellilinea caldifistulae*, *Lactivibrio alcoholicus*,
Acetomicrobium flavidum, *Spirochaeta thermophila*

G11: *Ferrimicrobium acidiphilum*, *Winkia neuui*, *Kocuria palustris*, *Saccharopolyspora*
rectivirgula

- Actinobacteria = 0.32
- Firmicutes = 0.693877
- Spirochaetes = 0.50
- All others = 0

⇒ **Weighted average (NJ) = 0.2452**

Weighted average calculation for UPGMA

Balanced UPGMA groups (12):

G1: *Calidithermus roseus*

G2: *Thermopolyspora flexuosa*, *Coleofasciculus chthonoplastes*, *Enterococcus florum*

G3: *Bacteroides fragilis*, *Bacteroides pyogenes*, *Lacinutrix algicola*, *Capnocytophaga cynodegm*

G4: *Simkania negevensis*, *Waddlia chondrophila*

G5: *Anaerolinea thermolimosa*, *Bellilinea caldifistulae*

G6: *Achromobacter denitrificans*, *Acetobacter aceti*, *Acidithiobacillus ferrooxidans*

G7: *Leptospira inadai*

G8: *Spirochaeta thermophila*

G9: *Lactivibrio alcoholicus*, *Acetomicrobium flavidum*

G10: *Ferrimicrobium acidiphilum*, *Winkia neuui*, *Saccharopolyspora rectivirgula*, *Kocuria*
palustris

G11: *Vallitalea guaymasensis*, *Staphylococcus aureus*, *Alkalibacterium pelagium*, *Enterococcus*
durans

G12: *Phascolarctobacterium faecium*, *Acidaminococcus fermentans*

G13: *Nostoc entophytum*

From this, Weighted average (UPGMA) = 0.245

Weighted average calculation for Morphology Tree

Balanced Morphology groups (12):

G1: *Coleofasciculus chthonoplastes*, *Nostoc entophytum*

G2 *Simkania negevensis*, *Waddlia chondrophila*

G3 *Acidaminococcus fermentans*, *Enterococcus durans*, *Staphylococcus aureus*, *Enterococcus florum*

G4 *Acetobacter aceti*, *Lacinutrix algicola*

G5 *Calidithermus roseus*, *Saccharopolyspora rectivirgula*

G6 *Kocuria palustris*, *Thermopolyspora flexuosa*

G7 *Leptospira inadai*

G8 *Acetomicrobium flavidum*, *Vallitalea guaymasensis*, *Spirochaeta thermophila*

G9: *Anaerolinea thermolimosa*, *Bellilinea caldifistulae*

G10: *Achromobacter denitrificans*, *Acetobacter aceti*, *Acidithiobacillus ferrooxidans*

G11: *Ferrimicrobium acidiphilum*, *Alkalibacterium pelagium*, *Deinococcus xinjiangensis*, *Winkia neuui*, *Capnocytophaga cynodegmi*

G12: *Bacteroides fragilis*, *Bacteroides pyogenes*, *Lactivibrio alcoholicus*, *Phascolarctobacterium faecium*

We can do the steps outlined above, and get Weighted average (Morphology) = 0.455

Interpretation:

ML remains best (lowest fragmentation of Phyla) as it has the lowest Gini Impurity. NJ and UPGMA both are worse than ML in terms of construction, and are quite similar in terms of weighted Gini Impurity. Morphology is by far the most scattered, with the highest Weighted Gini Index. Note that this method is not perfect and heavily depends on the grouping of clades. A rough algorithm was appropriated here for demonstrating the method, but more rigid algorithms to split the tree into clades is needed for reliable quantification. This method just looks at relationships at the tip, ignoring inter-clade relationships, and seems to suffice for this simple case. However for more complicated morphological trees, methods considering varying depths of the tree, and the relationship between clades will be ideal.